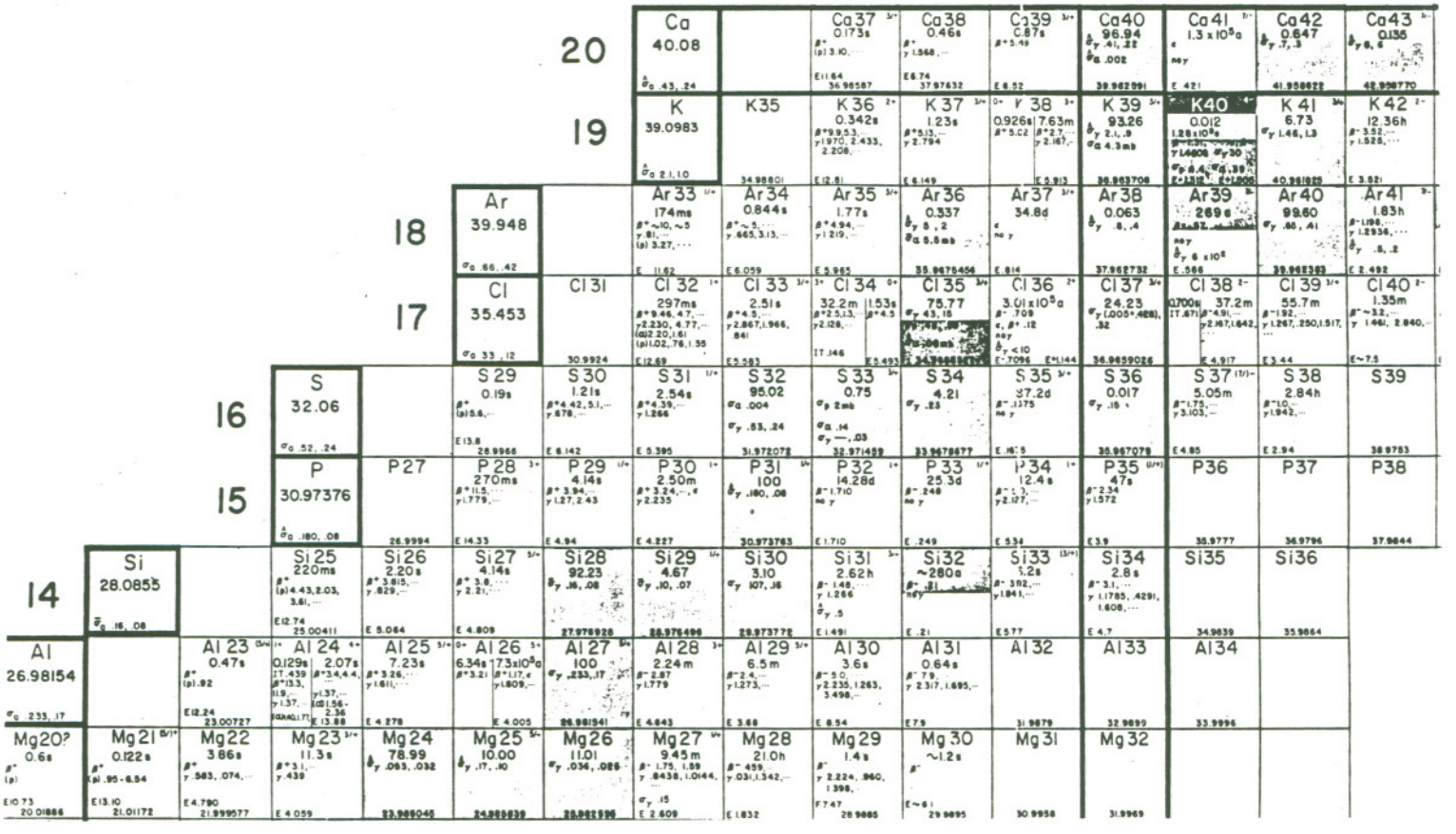
# ANALYTICAL CUMULATIVE EXAM

## January 16, 1993

NUCLEAR REACTIONS:

1. An important material in the semi-conductor industry is silicon doped with phosphorus. Using the excerpt from the Chart of the Nuclides shown below, design a procedure to produce this material involving neutron activation.




Show that the amount of phosphorous produced is given by $P = \sigma \emptyset N_t (1-e^{-\lambda t})$ where $\sigma$ is the relevant cross section for Si, $\emptyset$ the neutron flux ($10^{12}$ $cm^{-2}S^{-1}$), $N_t$ the target atom density, $\lambda$ the relevant decay constant and t the irradiation time. How long is it necessary to irradiate to produce 1% phosphorous in the Si?

2. Describe a method by which the amount of phosphorous being produced could be experimentally determined <u>during the irradiation</u>.

3. What type of detector is best suited to the purpose described in question 2?